



Barker
College

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Student Number

2024

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics Extension 2

Staff Involved:

- ARM - Mr Mallam
- JZT - Mrs Thomas
- BHC - Mr Carruthers*

PM THURSDAY 15TH AUGUST

45 copies

Topics Covered

Complex Numbers	Proof	Integration
Vectors	Mechanics	

General

Instructions:

- Write your Student Number at the top of this page and on all writing booklets
- Reading time - 10 minutes
- Working time - 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A separate reference booklet is provided
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale
- For questions in Section II, show relevant mathematical reasoning and / or calculations

Total marks:

100

Section I - 10 marks (pages 2 - 5)

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II - 90 marks (pages 6 - 12)

- Attempt Questions 11 - 16
- Allow about 2 hours and 45 minutes for this section
- Answer each question in a SEPARATE writing booklet

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

1 Which expression is equivalent to $\int \sin^3 x \, dx$?

- A. $\frac{1}{3} \cos^3 x + \cos x + c$
- B. $\frac{1}{3} \cos^3 x - \cos x + c$
- C. $\frac{1}{3} \sin^3 x + \sin x + c$
- D. $\frac{1}{3} \sin^3 x - \sin x + c$

2 A particle moves along a straight line with an acceleration of $\frac{2}{V} \text{ m/s}^2$, where V is the velocity at any instant in m/s.

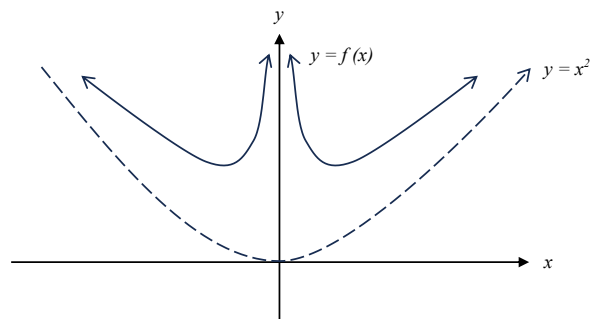
The initial velocity of the particle is -1 m/s . The particle will move to the:

- A. left, increasing in speed
- B. left, stop, then move to the right
- C. right, increasing in speed
- D. right, stop, then move to the left

3 What are the values of the real numbers p and q such that $1 - i$ is a root of the equation $z^3 + pz + q = 0$?

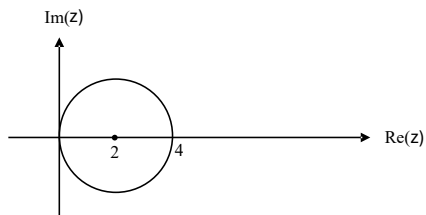
- A. $p = 2$ and $q = 4$
- B. $p = -2$ and $q = 4$
- C. $p = 2$ and $q = -4$
- D. $p = -2$ and $q = -4$

- 4 For the graph shown in the diagram below, a possible equation is:



- A. $y = \frac{x^3 + k}{x}, k > 0$
 B. $y = \frac{x^3 + k}{x}, k < 0$
 C. $y = \frac{x^4 + k}{x^2}, k > 0$
 D. $y = \frac{x^4 + k}{x^2}, k < 0$

- 5 Which of the following is the equation of the circle below?



- A. $(z - 2)(\bar{z} - 2i) = 4$
 B. $(z - 2)(\bar{z} - 2) = 4$
 C. $(z + 2)(\bar{z} - 2) = 4$
 D. $(z + 2i)(\bar{z} - 2i) = 4$

- 6 The position of a moving object is given by the cartesian coordinates $(3t, e^t)$, where t is the time in seconds.

Its acceleration is:

- A. constant in both magnitude and direction.
 B. constant in magnitude only.
 C. constant in direction only.
 D. constant in neither magnitude or direction.

- 7 Using an appropriate substitution, $\int e^{2x} \sqrt{e^x - 1} dx$ is equivalent to:

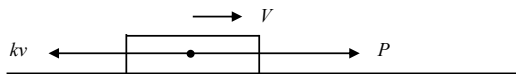
- A. $\int \left(u^{\frac{3}{2}} + u^{\frac{1}{2}} \right) du$
 B. $\int \left(u^{\frac{3}{2}} - u^{\frac{1}{2}} \right) du$
 C. $\int (u^3 + u) du$
 D. $\int (u^3 - u) du$

- 8 Let $z = \cos \theta + i \sin \theta$, where θ is acute.

The value of $\arg(z^2 - z)$ is:

- A. $\frac{\pi}{2} - \frac{\theta}{2}$
 B. $\frac{\pi}{2} + \frac{\theta}{2}$
 C. $\frac{\pi}{2} - \frac{3\theta}{2}$
 D. $\frac{\pi}{2} + \frac{3\theta}{2}$

- 9 The diagram below shows a box of mass m being pushed by a constant force P across a table.



The box experiences a resistive force due to friction which is proportional to its velocity, that is kv , and it is in the opposite direction.

There are no other forces acting on the box. By considering the forces on the box, its limiting velocity is given by:

- A. 0
- B. $\frac{P}{mk}$
- C. $\frac{P}{k}$
- D. $\sqrt{\frac{P}{k}}$
- 10 $J = \int_0^1 \sqrt{1-x^4} dx$
- $K = \int_0^1 \sqrt{1+x^4} dx$
- $L = \int_0^1 \sqrt{1-x^8} dx$

Which of the following statements is true for the definite integrals shown above?

- A. $J < L < 1 < K$
- B. $J < L < K < 1$
- C. $L < J < 1 < K$
- D. $L < J < K < 1$

End of Section I

Section II

90 marks

Attempt Questions 11 - 16

Allow about 2 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 - 16 your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) [START A NEW BOOKLET]

- (a) What is the cartesian equation of the line: 2

$$\underline{r} = \underline{i} + 2\underline{j} + \lambda(-3\underline{i} + 6\underline{j})?$$

- (b) Find r and θ such that: 2

$$e^{i\pi} - e^{i\frac{\pi}{2}} = re^{i\theta}$$

- (c) Find $\int x \sin 2x dx$ 3

- (d) For integer m , prove by contrapositive that if m^2 is not divisible by 4, then m is odd. 2

- (e) Prove that a particle with $x = 3 \cos^2 t$ is in simple harmonic motion. State its centre of motion and its amplitude. 3

- (f) Three vertices of a parallelogram are $O(0,0,0)$, $A(2,2,1)$ and $B(1,2,2)$. 3
Find all the possible positions of C , the fourth vertex.

Question 12 (15 marks) [START A NEW BOOKLET]

- (a) (i) Show that $\sqrt{2} - i$ is a root of the equation $z^3 - (\sqrt{2} - i)z^2 + 8z - 8\sqrt{2} + 8i = 0$ 1
- (ii) Find the other two solutions of the equation. 2
- (b) (i) Use de Moivre's Theorem and the Binomial Theorem to express $\cos 5\theta$ and $\sin 5\theta$ as powers of $\cos \theta$ and $\sin \theta$. 2
- (ii) Hence express $\tan 5\theta$ as a rational function of t , where $t = \tan \theta$. 2
- (iii) By considering the roots of $\tan 5\theta = 0$, deduce that: 2

$$\tan\left(\frac{\pi}{5}\right)\tan\left(\frac{2\pi}{5}\right)\tan\left(\frac{3\pi}{5}\right)\tan\left(\frac{4\pi}{5}\right) = 5$$

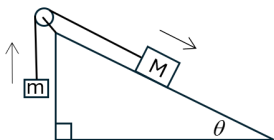
- (c) Any three dimensional vector can be expressed as scalar multiples of any three non-parallel vectors. Find λ, μ, ν such that $\underline{d} = \lambda\underline{a} + \mu\underline{b} + \nu\underline{c}$ where: 3

$$\underline{d} = 5\underline{i} + 5\underline{j} + 5\underline{k}, \quad \underline{a} = 2\underline{i} + 3\underline{j} + \underline{k}, \quad \underline{b} = \underline{i} - \underline{j} + 2\underline{k}, \quad \underline{c} = -\underline{i} + 2\underline{j} - \underline{k}$$

- (d) A block of mass M kg is connected to another block of mass m kg (where $M > m$) by a light inextensible string. The string is placed over a smooth pulley on an inclined plane, at an angle of θ to the horizontal, as shown. There is no friction between the surface of the inclined plane and the block with mass M kg.

Initially the system is held in equilibrium, but when released, the block with mass M kg begins to slide down the incline. Neglect the mass of the pulley.

The acceleration due to gravity is g . 3



Find the simplest expression for the tension, T , in the string.

Question 13 (15 marks) [START A NEW BOOKLET]

- (a) Let $I_n = \int_0^1 \frac{x^n}{(1+x)^2} dx$ for $n \geq 0$.
- (i) For $n \geq 2$, show that $I_n = \frac{1}{2(n-1)} - \frac{n}{n-1} I_{n-1}$ 4
- (ii) Hence find $\int_0^1 \frac{x^3}{(1+x)^2} dx$. 2

- (b) Find the shortest distance from the origin to the line through $A(1, 3, 1)$ and $B(0, 1, -1)$. 3

- (c) A body of unit mass is projected vertically upwards, under gravity, from the ground in a medium. This produces a resistance force of kv^2 , where v is the velocity and k is a positive constant. The acceleration due to gravity is g .

- (i) If the initial velocity is v_0 , prove that the maximum height, H , of the body above the ground is:

$$H = \frac{1}{2k} \log_e \left(1 + \frac{kv_0^2}{g} \right) \quad 3$$

- (ii) In a second projection vertically upwards of the body, it is noticed that the new maximum height reached is $2H$.

Show that the initial velocity of the second projection was $v_0(e^{2kH} + 1)^{\frac{1}{2}}$ 3

Question 14 (15 marks) **[START A NEW BOOKLET]**

- (a) (i) Prove that $\sqrt{10}$ is irrational. 2
- (ii) Hence prove that $\sqrt{2} + \sqrt{5}$ is irrational. 2
- (b) (i) Prove that $\sqrt{ab} \leq \frac{a+b}{2}$ where $a \geq 0$ and $b \geq 0$. 1
- (ii) Prove that $\sqrt{ab} \leq \sqrt{\frac{a^2+b^2}{2}}$ 1
- (iii) Prove that $\sqrt[4]{abcd} \leq \sqrt{\frac{a^2+b^2+c^2+d^2}{4}}$ 2
- (iv) If $1 \leq x \leq y$, show that $x(y-x+1) \geq y$ 2
- (v) Let n and k be positive integers with $1 \leq k \leq n$.
Prove that $\sqrt{n} \leq \sqrt{k(n-k+1)} \leq \frac{n+1}{2}$. 2
- (vi) For integers $n \geq 1$ prove that $(\sqrt{n})^n \leq n! \leq \left(\frac{n+1}{2}\right)^n$. 3

Question 15 (15 marks) **[START A NEW BOOKLET]**

- (a) (i) Expand and simplify $(x+1)(x^2-x+1)$. 1
- (ii) Hence evaluate $\int_0^\infty \frac{1}{1+x^3} dx$. 4
- (b) A stone is projected from a point on the ground and it just clears a fence d metres away.
The height of the fence is h metres.

The angle of projection is θ and the speed of projection is v m/s. Air resistance is negligible.
The displacement equations are $x = vt \cos \theta$ and $y = -\frac{1}{2}gt^2 + vt \sin \theta$.

(i) Show that $v^2 = \frac{gd^2 \sec^2 \theta}{2(d \tan \theta - h)}$ 2
- (ii) Show that the maximum height reached is $\frac{d^2 \tan^2 \theta}{4(d \tan \theta - h)}$ 2
- (iii) Show that the stone, when at its maximum height, will just clear the fence if $\tan \theta = \frac{2h}{d}$ 2
- (c) (i) Show that $\int_{-a}^a \frac{x^4}{1+e^x} dx = \int_{-a}^a \frac{x^4 e^x}{1+e^x} dx$. 2
- (ii) Hence, or otherwise, evaluate $\int_{-2}^2 \frac{x^4}{1+e^x} dx$ 2

Question 16 (15 marks) [START A NEW BOOKLET]

- (a) Relative to a fixed origin O , the horizontal unit vectors $\underline{i}$ and $\underline{j}$ are pointing due east and north respectively. A particle P , of mass 2kg, is moving under the action of a single constant force of $\underline{F}$ newtons.

When $t = 0$ seconds, the velocity of P is $(3\underline{i} - 5\underline{j})$ m/s and

when $t = 4$ seconds, the velocity of P is $(11\underline{i} + 7\underline{j})$ m/s.

You may assume that $\underline{v} = \underline{u} + \underline{a}t$, where $\underline{v}$ is the velocity at time t with initial velocity $\underline{u}$

- (i) Calculate the speed of the particle when $t = 0$. 1

- (ii) Determine the vector $\underline{F}$. 1

- (iii) Find the value of t at the moment the particle is moving due east. 1

- (b) Consider two spheres S_1 and S_2 .

$$S_1 : (x-1)^2 + (y+1)^2 + (z-2)^2 = 64$$

$$S_2 : (x+1)^2 + (y-3)^2 + (z+2)^2 = 4$$

- (i) Show that S_1 and S_2 are tangential (that is, they touch at only one point). 2

- (ii) Find the coordinates of the point of contact. 2

Question 16 continues on page 12

- (c) (i) Show that $e^{in\theta} - e^{-in\theta} = 2i \sin(n\theta)$ 1

- (ii) Consider the expression $S_n = 1 + e^{i\theta} + e^{i2\theta} + e^{i3\theta} + \dots + e^{in\theta}$

Show that $S_n = \frac{e^{i(n+1)\theta} - 1}{e^{i\theta} - 1}$ 1

- (iii) Show that $S_n = e^{i\left(\frac{n}{2}\right)\theta} \times \frac{\sin\left(\frac{n+1}{2}\right)\theta}{\sin\left(\frac{\theta}{2}\right)}$ 3

- (iv) Deduce that:

$$\sin\theta + \sin 2\theta + \dots + \sin n\theta = \frac{\sin\left(\frac{n}{2}\right)\theta \sin\left(\frac{n+1}{2}\right)\theta}{\sin\left(\frac{\theta}{2}\right)}$$
 3

End of Paper

Year 12 Extension 2 Mathematics Trial Examination 2024.

$$\begin{aligned} Q1, \int \sin^3 x \, dx &= \int (1 - \cos^2 x) \sin x \, dx \\ &= \int \sin x - \cos^2 x \sin x \, dx \\ &= -\cos x + \frac{1}{3} \cos^3 x + c \\ &= \frac{1}{3} \cos^3 x - \cos x + c \quad (B) \end{aligned}$$

Q2, The initial velocity is -1 m/s .
So it is moving to left.
Notice that $\ddot{x} = -2 \text{ m/s}$ initially.
So acceleration is in the same direction, so the particle will not stop, but move to the left increasing in speed. (A)

$$\begin{aligned} Q3, (1-i) \text{ is a root of } z^3 + pz + q &= 0, \text{ so} \\ (1-i)^3 + p(1-i) + q &= 0 \\ -2i(1-i) + p - ip + q &= 0 \\ -2i - 2 + p - ip + q &= 0 \\ (p+q-2) + i(-p-2) &= 0 \\ (p+q-2) - i(p+2) &= 0 \\ \therefore p &= -2 \\ q &= 4 \quad (B) \end{aligned}$$

Q4, The curve approaches the parabola as an asymptote from above. Hence $k > 0$.
Eliminate (B) and (D).
Notice that $y = f(x)$ is an even function.
(A) is not an even function.
(C) is an even function.
Hence (C).

$$\begin{aligned} Q5, \text{ The Cartesian equation of the circle is } (x-2)^2 + y^2 &= 4. \\ \text{Now if } (z-2)(\bar{z}-2) &= 4 \text{ then} \\ ((x-2)+iy)((x-2)-iy) &= 4 \\ (x-2)^2 - iy(x-2) + iy(x-2) + y^2 &= 4 \\ (x-2)^2 + y^2 &= 4 \\ \text{Hence (B).} \end{aligned}$$

$$\begin{aligned} Q6, \ddot{x} &= 3x & y &= e^t \\ \ddot{x} &= 3 & \dot{y} &= e^t \\ \ddot{x} &= 0 & \ddot{y} &= e^t \end{aligned}$$

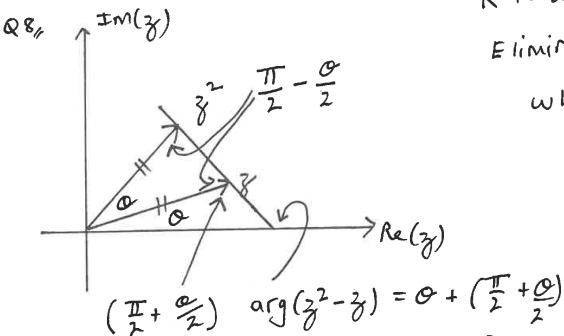
So no acceleration in x , only acceleration in y .
Acceleration is constant in direction only. Hence (C)

$$Q7, \int e^{2x} \sqrt{e^x - 1} \, dx$$

$$\begin{aligned} \text{Let } u &= e^x - 1 \\ du &= e^x \, dx \end{aligned}$$

$$\begin{aligned} I &= \int e^x \sqrt{e^x - 1} (e^x \, dx) \\ &= \int (u+1) \sqrt{u} \, du \\ &= \int u^{3/2} + u^{1/2} \, du \\ &= \frac{2}{5} u^{5/2} + \frac{2}{3} u^{3/2} + C \quad (A) \end{aligned}$$

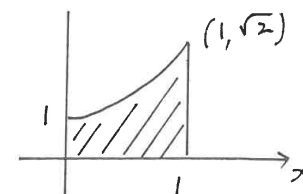
$$\begin{aligned} Q8, \arg(z^2 - z) &= \theta + \left(\frac{\pi}{2} + \frac{\theta}{2}\right) \\ &= \frac{\pi}{2} + \frac{3\theta}{2} \quad (D) \end{aligned}$$



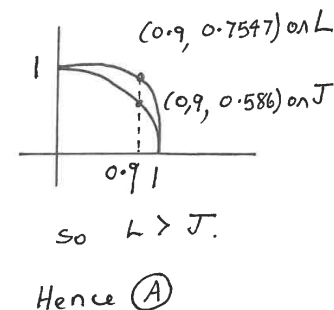
$$\begin{aligned} Q9, m\ddot{x} &= P - kV \\ \ddot{x} &= \frac{P}{m} - \frac{kV}{m} \\ \text{Solve } \ddot{x} &= 0 \\ \frac{P}{m} &= \frac{kV}{m} \\ V &= \frac{P}{k} \quad (C) \end{aligned}$$

$$Q10, \text{ Is } K < 1 \text{ or } K > 1?$$

The graph of $y = \sqrt{1+x^4}$ between $x=0$ and $x=1$ looks like this:



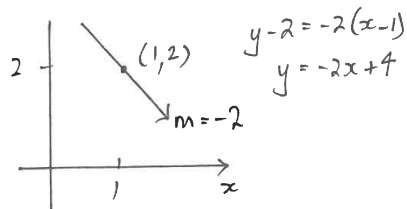
K is the shaded area, so $K > 1$.
Eliminate (B) and (D).
Which is bigger L or J ?



so $L > J$.
Hence (A)

Q11/ (a) The Cartesian equation of the line

$$\vec{r} = \vec{i} + 2\vec{j} + \lambda(-3\vec{i} + 6\vec{j})$$



(b) $e^{i\pi} - e^{i\frac{\pi}{2}}$

$$= \cos\pi + i\sin\pi - (\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$$

$$= -1 - i$$

$$= \sqrt{2} e^{-i\frac{3\pi}{4}}$$

Hence $r = \sqrt{2}$ and $\theta = -\frac{3\pi}{4}$

(c) $\int x \sin 2x \, dx$

$$= -\frac{1}{2} \cos 2x \cdot x - \int -\frac{1}{2} \cos 2x (1) \, dx$$

$$= -\frac{1}{2} x \cos 2x + \frac{1}{2} \int \cos 2x \, dx$$

$$= -\frac{1}{2} x \cos 2x + \frac{1}{4} \sin 2x + C$$

(d) The contrapositive is:

"If m is not odd, then m^2 is divisible by 4."

Suppose that m is even.

Let $m = 2k$ (k an integer).

$$\text{Then } m^2 = 4k^2,$$

which is divisible by 4.

So if m is even, then

m^2 is divisible by 4.

Hence, if m^2 is not

divisible by 4, then

m is odd.

(e) $x = 3\cos^2 t$

$$\dot{x} = -6 \cos t \sin t$$

$$= -3 \sin 2t$$

$$\ddot{x} = -6 \cos 2t$$

But $\cos 2t = 2\cos^2 t - 1$ so

$$\ddot{x} = -6(2\cos^2 t - 1)$$

$$= -12 \cos^2 t + 6$$

$$= -4(3\cos^2 t) + 6$$

$$= -4x + 6$$

$$= -4(x - \frac{3}{2})$$

which is of the form

$$\ddot{x} = -n^2(x - a)$$

The centre of motion is

$$x = \frac{3}{2} \text{ and the amplitude is } \frac{3}{2}.$$

(f) There are 3 possible positions for the fourth vertex, D,

to make a parallelogram:

$$\vec{OD} = \vec{OA} + \vec{OB}$$

$$\vec{OD} = \vec{OA} - \vec{OB}$$

$$\vec{OD} = \vec{OB} - \vec{OA}$$

Hence $\vec{OD} = (2, 2, 1) + (1, 2, 2) = (3, 4, 3)$ or

$$\vec{OD} = (2, 2, 1) - (1, 2, 2) = (1, 0, -1)$$
 or

$$\vec{OD} = (1, 2, 2) - (2, 2, 1) = (-1, 0, 1).$$

Q12//

(a) (i) By substitution,

$$(\sqrt{2}-i)^3 - (\sqrt{2}-i)(\sqrt{2}-i)^3 + 8(\sqrt{2}-i) - 8\sqrt{2} + 8i$$

$$= 0 + 8\sqrt{2} - 8i - 8\sqrt{2} + 8i$$

$$= 0$$

Hence $\sqrt{2}-i$ is a root of the equation.

(ii) The sum of the roots is $-\frac{b}{a} = \sqrt{2}-i$.

$$\text{Hence } \alpha + \beta + \sqrt{2}-i = \sqrt{2}-i$$

$$\alpha + \beta = 0$$

$$\alpha = -\beta$$

and the product of the roots is $-\frac{d}{a} = 8\sqrt{2}-8i$

$$\text{Hence } \alpha\beta(\sqrt{2}-i) = 8\sqrt{2}-8i$$

$$\alpha\beta = \frac{8\sqrt{2}-8i}{\sqrt{2}-i} \times \frac{\sqrt{2}+i}{\sqrt{2}+i}$$

$$\alpha(-\alpha) = \frac{16-8\sqrt{2}i+8\sqrt{2}i+8}{3}$$

$$-\alpha^2 = 8$$

$$\alpha^2 = -8$$

$$\alpha = \pm 2\sqrt{2}i$$

If $\alpha = 2\sqrt{2}i$ then $\beta = -2\sqrt{2}i$, or vice versa.

(b) Now $(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$ by de Moivre.

(i) By the Binomial Theorem

$$(\cos \theta + i \sin \theta)^5$$

$$= \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (i \sin \theta)^2 + 10 \cos^2 \theta (i \sin \theta)^3 + 5 \cos \theta (i \sin \theta)^4 + (i \sin \theta)^5$$

$$= \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) - 10 \cos^3 \theta \sin^2 \theta - 10 \cos^2 \theta (i \sin^3 \theta) + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta$$

$$= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta + i (5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta)$$

By equating Real parts

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

and by equating imaginary parts

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

$$(ii) \tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta}$$

$$= \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

(Now divide top and bottom by $\cos^5 \theta$)

$$= \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - \tan^2 \theta + 5 \tan^4 \theta}$$

$$= \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4}, \text{ where } t = \tan \theta.$$

(iii) Consider solving $\tan 5\theta = 0$.

$$\text{Then } 5t - 10t^3 + t^5 = 0$$

$$t(5 - 10t^2 + t^4) = 0$$

$$\text{So } t = 0 \text{ or } t^4 - 10t^2 + 5 = 0$$

From $\tan 5\theta = 0$

$$5\theta = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$\theta = 0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}$$

Hence the four roots of $t^4 - 10t^2 + 5 = 0$
are $\tan \frac{\pi}{5}, \tan \frac{2\pi}{5}, \tan \frac{3\pi}{5}, \tan \frac{4\pi}{5}$

and the product of the roots is $\frac{e}{a} = 5$

$$\text{Hence } \tan \frac{\pi}{5} \times \tan \frac{2\pi}{5} \times \tan \frac{3\pi}{5} \times \tan \frac{4\pi}{5} = 5,$$

as required.

$$(c) \begin{pmatrix} 5 \\ 5 \\ 5 \end{pmatrix} = \lambda \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} + \nu \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} 5 \\ 5 \\ 5 \end{pmatrix} = \begin{pmatrix} 2\lambda + \mu - \nu \\ 3\lambda - \mu + 2\nu \\ \lambda + 2\mu - \nu \end{pmatrix}$$

$$2\lambda + \mu - \nu = 5 \quad \text{--- (1)}$$

$$3\lambda - \mu + 2\nu = 5 \quad \text{--- (2)}$$

$$\lambda + 2\mu - \nu = 5 \quad \text{--- (3)}$$

$$\text{(1) + (2) gives } 5\lambda + \nu = 10 \quad \text{--- (4)}$$

$$\text{(3) + 2 \times (2) gives } 7\lambda + 3\nu = 15 \quad \text{--- (5)}$$

3 \times (1) gives

$$15\lambda + 3\nu = 30 \quad \text{--- (6)}$$

(6) - (5) gives

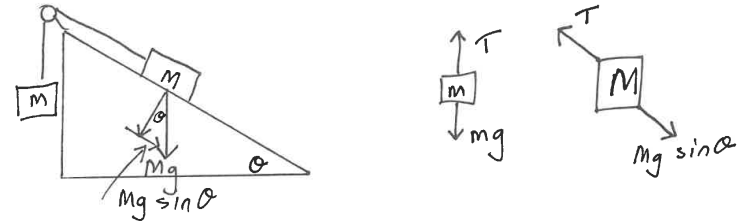
$$8\lambda = 15$$

$$\therefore \lambda = \frac{15}{8}$$

$$\therefore \nu = \frac{5}{8}$$

$$\therefore \mu = \frac{15}{8}$$

(d)



For block M,

$$Mg \sin \theta - T = M\ddot{x} \quad \text{--- (1)}$$

For block m,

$$T - mg = m\ddot{x} \quad \text{--- (2)}$$

Add (1) and (2) to get

$$Mg \sin \theta - mg = \ddot{x}(m + M)$$

$$\ddot{x} = \frac{Mg \sin \theta - mg}{m + M}$$

Substitute this into (2) to get

$$T = mg + m \left(\frac{Mg \sin \theta - mg}{m + M} \right)$$

$$= \frac{mg(m + M) + mMg \sin \theta - m^2g}{m + M}$$

$$= \frac{mMg + mMg \sin \theta}{m + M}$$

$$= \frac{mMg(1 + \sin \theta)}{m + M}$$

Q13, (a) (i)

$$\begin{aligned}
 I_n &= \int_0^1 \frac{x^n}{(1+x)^2} dx \\
 &= \int_0^1 x^n (1+x)^{-2} dx \\
 &= \left[-(1+x)^{-1} x^n \right]_0^1 - \int_0^1 -(1+x)^{-1} n x^{n-1} dx \\
 &= -\frac{1}{2} + n \int_0^1 \frac{x^{n-1}}{(1+x)} dx \\
 &= -\frac{1}{2} + n \left[\int_0^1 \frac{x^{n-1} (1+x)}{(1+x)^2} dx \right] \\
 &= -\frac{1}{2} + n \left[\int_0^1 \frac{x^{n-1}}{(1+x)^2} dx + \int_0^1 \frac{x^n}{(1+x)^2} dx \right] \\
 &= -\frac{1}{2} + n I_{n-1} + n I_n
 \end{aligned}$$

$$I_n - n I_n = -\frac{1}{2} + n I_{n-1}$$

$$I_n (1-n) = -\frac{1}{2} + n I_{n-1}$$

$$I_n = \frac{-1}{2(1-n)} + \frac{n}{1-n} I_{n-1}$$

$$= \frac{1}{2(n-1)} - \frac{n}{n-1} I_{n-1}, \text{ as required.}$$

$$(ii) \int_0^1 \frac{x^3}{(1+x)^2} dx = I_3$$

$$I_3 = \frac{1}{4} - \frac{3}{2} I_2$$

$$I_2 = \frac{1}{2} - 2 I_1$$

$$I_1 = \int_0^1 \frac{x}{(1+x)^2} dx$$

$$\text{let } u = 1+x \\ du = dx$$

$$\text{when } x=1, u=2$$

$$\text{when } x=0, u=1$$

$$I_1 = \int_1^2 \frac{u-1}{u^2} du$$

$$= \int_1^2 \frac{1}{u} - \frac{1}{u^2} du$$

$$= \left[\ln u + \frac{1}{u} \right]_1^2$$

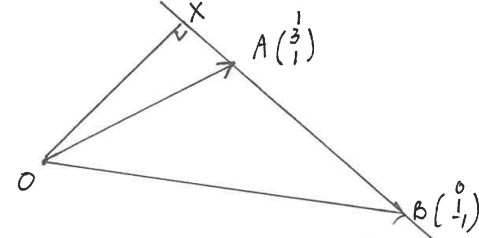
$$= \ln 2 - \frac{1}{2}$$

$$I_2 = \frac{3}{2} - 2 \ln 2$$

$$I_3 = -2 + 3 \ln 2$$

Q13, (b) Now $A(1, 3, 1)$ and $B(0, 1, -1)$.

$$\begin{aligned}
 \text{so } \vec{AB} &= \vec{OB} - \vec{OA} \\
 &= \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix}
 \end{aligned}$$



The shortest vector from the origin to the line $\vec{AB}$ is the vector $\vec{OX}$ where $\vec{OX} = \vec{OA} + \vec{AX}$ and $\vec{AX}$ is the projection of $\vec{AO}$ onto the line AB .

$$\begin{aligned}
 \text{Now } \text{proj}_{\vec{AB}} \vec{AO} &= \frac{\begin{pmatrix} -1 \\ -3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix}}{1^2 + 2^2 + 2^2} \times \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix}
 \end{aligned}$$

$$= \frac{1+6+2}{9} \times \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix}$$

$$\vec{OX} = \vec{OA} + \vec{AX}$$

$$= \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$\text{and } |\vec{OX}| = \sqrt{2} \text{ units}$$

$$(c) \text{ (i) } v \frac{dv}{dx} = -g - kv^2$$

$$\frac{dv}{dx} = -\frac{(g + kv^2)}{v}$$

$$\frac{dx}{dv} = \frac{-v}{g + kv^2}$$

$$\text{Max height} = \int_{v_0}^0 \frac{-v dv}{g + kv^2}$$

$$H = -\frac{1}{2k} \int_{v_0}^0 \frac{2kv}{g + kv^2} dv$$

$$= \frac{1}{2k} \int_0^{v_0} \frac{2kv}{g + kv^2} dv$$

$$= \frac{1}{2k} \left[\ln(g + kv^2) \right]_0^{v_0}$$

$$= \frac{1}{2k} \ln(g + kv_0^2) - \frac{1}{2k} \ln(g)$$

$$= \frac{1}{2k} \ln\left(\frac{g + kv_0^2}{g}\right) = \frac{1}{2k} \ln\left(1 + \frac{kv_0^2}{g}\right)$$

as required.

(ii) Let the new initial velocity be v_1 .

$$2H = \frac{1}{2k} \ln\left(1 + \frac{kv_1^2}{g}\right)$$

$$4kH = \ln\left(1 + \frac{kv_1^2}{g}\right)$$

$$e^{4kH} = 1 + \frac{kv_1^2}{g}$$

$$v_1^2 = \frac{g}{k} (e^{4kH} - 1)$$

$$\text{So } v_1 = \sqrt{\frac{g}{k} (e^{4kH} - 1)}$$

$$\text{Now } v_0 = \sqrt{\frac{g}{k} (e^{2kH} - 1)}$$

$$\begin{aligned} \text{So } v_1 &= \sqrt{\frac{g}{k} (e^{2kH} - 1) (e^{2kH} + 1)} \\ &= \sqrt{\frac{g}{k} (e^{2kH} - 1)} \times \sqrt{e^{2kH} + 1} \\ &= v_0 \times \sqrt{e^{2kH} + 1} \\ &= v_0 (e^{2kH} + 1)^{\frac{1}{2}}, \\ &\text{as required.} \end{aligned}$$

Q14 (a)

(i) Proof by contradiction:

Suppose that $\sqrt{10}$ is a rational number and can be written in the form $\sqrt{10} = \frac{p}{q}$ where p and q are integers with no common factors.

$$\text{Then } 10 = \frac{p^2}{q^2}$$

$$p^2 = 10q^2 \quad \text{--- (1)}$$

so p^2 is even

so p is even

let $p = 2m$. Sub into (1):

$$(2m)^2 = 10q^2$$

$$4m^2 = 10q^2$$

$$2m^2 = 5q^2$$

$5q^2$ is even

so q^2 is even

so q is even

Contradiction!

p and q cannot both be even as they have no common factors.

Hence $\sqrt{10}$ is not a rational number.

$\sqrt{10}$ is irrational.

(ii) Proof by contradiction:

Suppose that $\sqrt{2} + \sqrt{5}$ is a rational number.

Let $\sqrt{2} + \sqrt{5} = \frac{p}{q}$ where

p and q are integers.

$$\text{Then } (\sqrt{2} + \sqrt{5})^2 = \frac{p^2}{q^2}$$

$$7 + 2\sqrt{10} = \frac{p^2}{q^2}$$

$$2\sqrt{10} = \frac{p^2 - 7q^2}{q^2}$$

$$\sqrt{10} = \frac{p^2 - 7q^2}{2q^2}$$

Contradiction!

The RHS is an integer $(p^2 - 7q^2)$ divided by another integer $(2q^2)$, and therefore rational.

But the LHS is $\sqrt{10}$ which we proved in (i) to be irrational.

Hence $\sqrt{2} + \sqrt{5}$ is an irrational number.

Q14 (b)

(i) Now $(a-b)^2 \geq 0$

$$a^2 - 2ab + b^2 \geq 0$$

$$a^2 + b^2 \geq 2ab$$

$$ab \leq \frac{a^2 + b^2}{2}$$

Sub $\sqrt{a}$ for a and $\sqrt{b}$ for b

$$\sqrt{a} \cdot \sqrt{b} \leq \frac{(\sqrt{a})^2 + (\sqrt{b})^2}{2}$$

$$\sqrt{ab} \leq \frac{a+b}{2}$$

(ii) Now $ab \leq \frac{a^2 + b^2}{2}$ from (i) above

$$\therefore \sqrt{ab} \leq \sqrt{\frac{a^2 + b^2}{2}}$$

(iii) From (ii) above

$$\sqrt{ab} \leq \sqrt{\frac{a^2 + b^2}{2}}$$

$$\text{So } \sqrt{cd} \leq \sqrt{\frac{c^2 + d^2}{2}}$$

$$\sqrt{\sqrt{ab} \times \sqrt{cd}} \leq \sqrt{\frac{\left(\sqrt{\frac{a^2 + b^2}{2}}\right)^2 + \left(\sqrt{\frac{c^2 + d^2}{2}}\right)^2}{2}}$$

$$4\sqrt{abcd} \leq \sqrt{\frac{a^2 + b^2 + c^2 + d^2}{4}}$$

Alternatively:

$$ab \leq \frac{a^2 + b^2}{2}$$

$$\text{and } cd \leq \frac{c^2 + d^2}{2}$$

$$\therefore ab + cd \leq \frac{a^2 + b^2}{2} + \frac{c^2 + d^2}{2}$$

$$\leq \frac{a^2 + b^2 + c^2 + d^2}{2} \quad \textcircled{2}$$

$$\sqrt{abcd} = \sqrt{(ab)(cd)}$$

$$\leq \frac{ab + cd}{2} \quad (\text{by given inequality})$$

$$\leq \frac{a^2 + b^2 + c^2 + d^2}{4} \quad \text{from } \textcircled{2}$$

Taking square roots:

$$\sqrt[4]{abcd} \leq \sqrt{\frac{a^2 + b^2 + c^2 + d^2}{4}}$$

(iv) If $1 \leq x \leq y$, show that

$$x(y-x+1) \geq y$$

Now $y \geq x$

$$\text{so } y(x-1) \geq x(x-1)$$

because $x \geq 1$

$$\text{so } xy - y \geq x^2 - x$$

$$xy - x^2 + x \geq y$$

$$\therefore x(y-x+1) \geq y$$

Alternatively:

LHS - RHS

$$= xy - x^2 + x - y$$

$$= y(x-1) - x(x-1)$$

$$= (y-x)(x-1)$$

↑ this is positive or zero

this is positive or zero

$$\therefore \text{LHS} - \text{RHS} \geq 0$$

$$\therefore x(y-x+1) \geq y$$

(v) Let n and k be positive integers with $1 \leq k \leq n$.
Prove that $\sqrt{n} \leq \sqrt{k(n-k+1)} \leq \frac{n+1}{2}$.

$$\text{From (iv) } x(y-x+1) \geq y$$

Substitute $x=k$ and $y=n$, then

$$k(n-k+1) \geq n$$

$$\therefore \sqrt{k(n-k+1)} \geq \sqrt{n}$$

$$\therefore \sqrt{n} \leq \sqrt{k(n-k+1)} \quad \text{--- } \textcircled{2}$$

$$\text{From (i) } \sqrt{ab} \leq \frac{a+b}{2} \quad \text{so}$$

$$\sqrt{k(n-k+1)} \leq \frac{k + (n-k+1)}{2}$$

$$\text{i.e., } \sqrt{k(n-k+1)} \leq \frac{n+1}{2}$$

$$\text{Hence } \sqrt{n} \leq \sqrt{k(n-k+1)} \leq \frac{n+1}{2}$$

(vi) For integers $n \geq 1$, to prove that

$$(\sqrt{n})^n \leq n! \leq \left(\frac{n+1}{2}\right)^n$$

Substitute $k=1$ into (i) above gives $\sqrt{n} \leq \sqrt{1(n)} \leq \frac{n+1}{2}$

$$\text{" } k=2 \text{ " " " " " } \sqrt{n} \leq \sqrt{2(n-1)} \leq \frac{n+1}{2}$$

$$\text{" } k=3 \text{ " " " " " } \sqrt{n} \leq \sqrt{3(n-2)} \leq \frac{n+1}{2}$$

$$\text{etc}$$

$$\text{" } k=n \text{ " " " " " } \sqrt{n} \leq \sqrt{n} \leq \frac{n+1}{2}$$

Now multiply all the inequalities above together to get

$$(\sqrt{n})^n \leq \sqrt{(n(n-1)(n-2) \dots 3 \times 2 \times 1)^2} \leq \left(\frac{n+1}{2}\right)^n$$

$$\text{Hence } (\sqrt{n})^n \leq n! \leq \left(\frac{n+1}{2}\right)^n, \text{ as required.}$$

Q15 (a)

$$\begin{aligned} \text{(i)} \quad & (x+1)(x^2-x+1) \\ &= x^3 - x^2 + x + x^2 - x + 1 \\ &= x^3 + 1 \end{aligned}$$

$$\text{(ii)} \quad \int_0^{\infty} \frac{1}{1+x^3} dx$$

$$\text{Let } \frac{1}{1+x^3} = \frac{A}{1+x} + \frac{Bx+C}{x^2-x+1}$$

$$1 = A(x^2-x+1) + (Bx+C)(1+x)$$

$$\text{Let } x = -1, \quad 1 = 3A \quad \text{so } A = \frac{1}{3}$$

$$\text{Let } x = 0, \quad 1 = A + C \quad \text{so } C = \frac{2}{3}$$

$$\begin{aligned} \text{Let } x = 1, \quad & 1 = A + 2(B+C) \\ & 1 = \frac{1}{3} + 2B + \frac{4}{3} \quad \text{so } B = -\frac{1}{3} \end{aligned}$$

$$\therefore \int_0^{\infty} \frac{dx}{1+x^3} = \int_0^{\infty} \frac{\frac{1}{3}}{1+x} + \frac{-\frac{1}{3}x + \frac{2}{3}}{x^2-x+1} dx$$

$$= \left[\frac{1}{3} \ln|1+x| \right]_0^{\infty} - \frac{1}{3} \int_0^{\infty} \frac{x-2}{x^2-x+1} dx$$

$$= \left[\frac{1}{3} \ln|1+x| \right]_0^{\infty} - \frac{1}{3} \int_0^{\infty} \frac{\frac{1}{2}(2x-1) - \frac{3}{2}}{x^2-x+1} dx$$

$$= \left[\frac{1}{3} \ln|1+x| \right]_0^{\infty} - \frac{1}{6} \int_0^{\infty} \frac{2x-1}{x^2-x+1} dx + \frac{1}{2} \int_0^{\infty} \frac{dx}{(x-\frac{1}{2})^2 + \frac{3}{4}}$$

$$= \left[\frac{1}{6} \ln(1+x)^2 \right]_0^{\infty} - \left[\frac{1}{6} \ln|x^2-x+1| \right]_0^{\infty} + \frac{1}{2} \cdot \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x-\frac{1}{2}}{\sqrt{3}/2} \right) \Bigg|_0^{\infty}$$

$$= \left[\frac{1}{6} \ln \frac{(x+1)^2}{x^2-x+1} \right]_0^{\infty} + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x-1}{\sqrt{3}} \right) \Bigg|_0^{\infty}$$

↑

This term approaches zero.

$$= 0 + \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{\infty-1}{\sqrt{3}}\right) - \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$$

$$= \frac{1}{\sqrt{3}} \times \frac{\pi}{2} - \frac{1}{\sqrt{3}} \left(-\frac{\pi}{6}\right)$$

$$= \frac{\pi}{2\sqrt{3}} + \frac{\pi}{6\sqrt{3}}$$

$$= \frac{4\pi}{6\sqrt{3}}$$

$$= \frac{2\pi}{3\sqrt{3}}$$

(b) Now $t = \frac{x}{V \cos \theta}$

(i) $y = -\frac{1}{2}g \left(\frac{x}{V \cos \theta}\right)^2 + V \left(\frac{x}{V \cos \theta}\right) \sin \theta$

$$y = \frac{-gx^2}{2V^2} \sec^2 \theta + x \tan \theta$$

Let $x = d$ and $y = h$, then

$$h = \frac{-gd^2}{2V^2} \sec^2 \theta + d \tan \theta$$

$$\frac{gd^2}{2V^2} \sec^2 \theta = d \tan \theta - h$$

$$V^2 = \frac{gd^2 \sec^2 \theta}{2(d \tan \theta - h)}$$

as required.

(iii) $h = \frac{d^2 \tan^2 \theta}{4(d \tan \theta - h)}$

$$4dh \tan \theta - 4h^2 = d^2 \tan^2 \theta$$

$$d^2 \tan^2 \theta - 4dh \tan \theta + 4h^2 = 0$$

$$(d \tan \theta - 2h)^2 = 0$$

(ii) Max height occurs on the axis of symmetry.

$$t = \frac{V \sin \theta}{g}$$

$$y_{\max} = -\frac{1}{2}g \left(\frac{V \sin \theta}{g}\right)^2 +$$

$$V \left(\frac{V \sin \theta}{g}\right) \sin \theta$$

$$= -\frac{V^2 \sin^2 \theta}{2g} + \frac{V^2 \sin^2 \theta}{g}$$

$$= \frac{V^2 \sin^2 \theta}{2g}$$

$$= V^2 \left(\frac{\sin^2 \theta}{2g}\right)$$

$$= \frac{gd^2 \sec^2 \theta}{2(d \tan \theta - h)} \times \frac{\sin^2 \theta}{2g}$$

$$= \frac{d^2 \tan^2 \theta}{4(d \tan \theta - h)}$$

as required.

Q15/ (c)

(i) To show that

$$I = \int_{-a}^a \frac{x^4 dx}{1+e^x} = \int_{-a}^a \frac{x^4 e^x dx}{1+e^x}$$

Let $x = -u$

$$dx = -du$$

When $x = a$, $u = -a$

When $x = -a$, $u = a$

$$I = \int_a^{-a} \frac{(-u)^4 (-du)}{1+e^{-u}}$$

$$= \int_{-a}^a \frac{u^4 du}{1+e^{-u}}$$

$$= \int_{-a}^a \frac{u^4 e^u du}{1+e^u}$$

$$= \int_{-a}^a \frac{x^4 e^x dx}{1+e^x}$$

(change of variable)

(ii) $\int_{-2}^2 \frac{x^4 dx}{1+e^x} = \int_{-2}^2 \frac{x^4 e^x dx}{1+e^x}$

$$\therefore 2I = \int_{-2}^2 \frac{x^4 (1+e^x)}{(1+e^x)} dx$$

$$= \int_{-2}^2 x^4 dx$$

$$= \left[\frac{1}{5}x^5\right]_{-2}^2$$

$$= 64/5$$

$$\therefore I = \frac{32}{5}$$

Q16/ (a) (i) When $t=0$, $\underline{v} = 3\hat{i} - 5\hat{j}$ m/s

Hence speed $= |\underline{v}| = |3\hat{i} - 5\hat{j}|$
 $= \sqrt{3^2 + 5^2}$
 $= \sqrt{34}$ m/s

(ii) Using $\underline{v} = \underline{u} + \underline{a}t$

$11\hat{i} + 7\hat{j} = (3\hat{i} - 5\hat{j}) + \underline{a} \times 4$

$8\hat{i} + 12\hat{j} = 4\underline{a}$

$\underline{a} = 2\hat{i} + 3\hat{j}$ m/s²

Hence $F = m\underline{a}$

$= 2(2\hat{i} + 3\hat{j})$

$= 4\hat{i} + 6\hat{j}$ N

(iii) $\underline{v} = \underline{u} + \underline{a}t$

$= (3\hat{i} - 5\hat{j}) + (2\hat{i} + 3\hat{j})t$

$= (2t+3)\hat{i} + (3t-5)\hat{j}$

When the particle is moving east, $\hat{j} = 0$ so

$3t - 5 = 0$

$t = \frac{5}{3}$ seconds.

(b)

(i) The centre of S_1 is at $C_1(1, -1, 2)$ and its radius is 8.

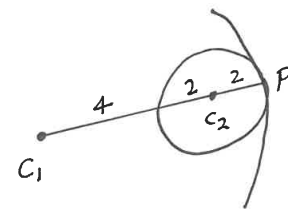
The centre of S_2 is at $C_2(-1, 3, -2)$ and its radius is 2.

The distance between C_1 and C_2 is

$d = \sqrt{(1-(-1))^2 + (-1-3)^2 + (2-(-2))^2} = \sqrt{36} = 6$ units

Notice that S_2 is inside S_1 .

The two spheres touch at P.



$C_1C_2 = 6$

$C_2P = \frac{2}{8}$

The two spheres are tangential.

(ii) To find the coordinates of the point of contact, we need to extend the direction vector $\overrightarrow{C_1C_2}$ by $\frac{1}{3}$.

Consider the $\hat{i}$ component: from 1 to -1 extended by $\frac{1}{3}$ takes us to $-\frac{5}{3}$.

Consider the $\hat{j}$ component: from -1 to 3 extended by $\frac{1}{3}$ takes us to $\frac{13}{3}$.

Consider the $\hat{k}$ component: from 2 to -2 extended by $\frac{1}{3}$ takes us to $-\frac{10}{3}$.

Hence $P = \left(-\frac{5}{3}, \frac{13}{3}, -\frac{10}{3}\right)$.

$$\begin{aligned}
 \text{(c) (i)} \quad e^{in\theta} - e^{-in\theta} &= \cos(n\theta) + i\sin(n\theta) - [\cos(-n\theta) + i\sin(-n\theta)] \\
 &= \cos(n\theta) + i\sin(n\theta) - [\cos(n\theta) - i\sin(n\theta)] \\
 &= \cos(n\theta) + i\sin(n\theta) - \cos(n\theta) + i\sin(n\theta) \\
 &= 2i\sin(n\theta)
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad S_n &= 1 + e^{i\theta} + e^{i2\theta} + \dots + e^{in\theta} \\
 &= 1 \left[\frac{(e^{i\theta})^{n+1} - 1}{e^{i\theta} - 1} \right] \\
 &= \frac{e^{i(n+1)\theta} - 1}{e^{i\theta} - 1}, \text{ as required}
 \end{aligned}$$

$$\text{(iii) Now } S_p = \frac{e^{i(n+1)\theta} - 1}{e^{i\theta} - 1} \text{ from above.}$$

$$\begin{aligned}
 S_n &= \frac{e^{i(\frac{n+1}{2})\theta} [e^{i(\frac{n+1}{2})\theta} - e^{-i(\frac{n+1}{2})\theta}]}{e^{i(\frac{\theta}{2})} [e^{i(\frac{\theta}{2})} - e^{-i(\frac{\theta}{2})}]} \\
 &= \frac{e^{i(\frac{n+1}{2})\theta}}{e^{i(\frac{\theta}{2})}} \times \frac{2i \sin(\frac{n+1}{2}\theta)}{2i \sin(\frac{\theta}{2})} \\
 &= e^{i(\frac{n}{2})\theta} \times \frac{\sin(\frac{n+1}{2}\theta)}{\sin(\frac{\theta}{2})}, \text{ as required.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv) Now } \sin\theta + \sin 2\theta + \dots + \sin n\theta &= \text{Im} (1 + e^{i\theta} + e^{i2\theta} + e^{i3\theta} + \dots + e^{in\theta})
 \end{aligned}$$

$$\begin{aligned}
 &= \text{Im} \left(e^{i(\frac{n}{2})\theta} \times \frac{\sin(\frac{n+1}{2}\theta)}{\sin(\frac{\theta}{2})} \right) \\
 &= \text{Im} \left[(\cos(\frac{n}{2}\theta) + i\sin(\frac{n}{2}\theta)) \times \frac{\sin(\frac{n+1}{2}\theta)}{\sin(\frac{\theta}{2})} \right]
 \end{aligned}$$

$$\begin{aligned}
 &= \text{Im} \left[\cos(\frac{n}{2}\theta) \frac{\sin(\frac{n+1}{2}\theta)}{\sin(\frac{\theta}{2})} + i \frac{\sin(\frac{n}{2}\theta) \sin(\frac{n+1}{2}\theta)}{\sin(\frac{\theta}{2})} \right] \\
 &= \sin(\frac{n}{2}\theta) \times \frac{\sin(\frac{n+1}{2}\theta)}{\sin(\frac{\theta}{2})}, \text{ as required.}
 \end{aligned}$$

End of Paper